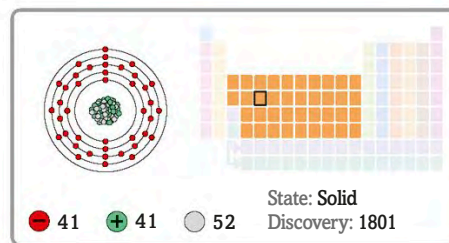


41
Nb

Niobium



Forms

This dark, dense ore has a light metallic shine.

Columbite



This greyish pure metal is soft.

Rods of pure niobium refined in a laboratory



Columbite is named after the country of **Columbia**.

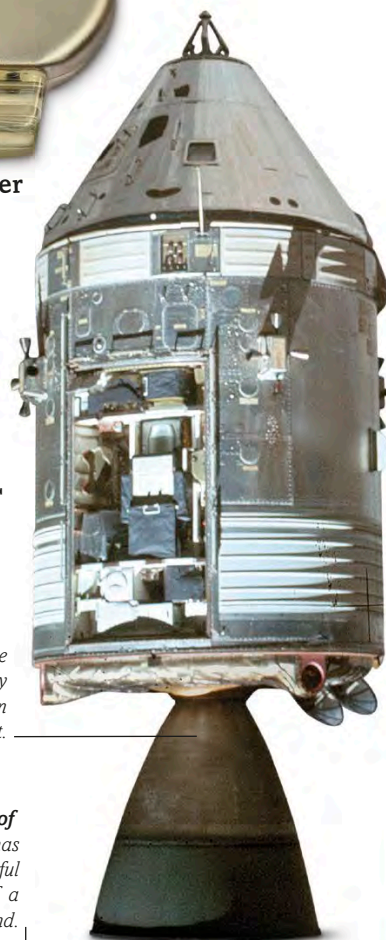
Uses



Pacemaker

The niobium case houses a battery that regulates the beating of a human heart.

Command Module from the Apollo 15 mission



This nozzle made of a niobium alloy kept its shape even when very hot.

This pair of spectacles has thin, powerful lenses made of a niobium compound.



Spectacle lenses

Niobium is so similar to the metal tantalum that the two were wrongly thought to be the same element for almost 40 years. The mineral **columbite** is the main source of this shiny metal. Niobium is not found naturally in its **pure form**. When extracted, it has many uses. As the element

does not react adversely in the human body, it is used in implants, such as **pacemakers**. Niobium also does not expand when hot, so it is used to make parts of rockets, such as the one on the **Command Module** from NASA's Apollo 15 spacecraft that went to the Moon in 1971.